

$$\begin{aligned}
T^{-1} &: X_0 \times B \rightarrow \mathcal{X} \\
T^{-1}(x, t) &= (z(x, t), t) \\
\pi(z(x, t), t) &= x
\end{aligned}$$

Denote the vector field of type-(1,0) on $\mathcal{X}$ as $\nu = v - iJ(v)$ where J is the complex structure. Due to the good trivialization, we have both of v and $J(v)$ tangent to a complex submanifold being projected onto a point. Hence $\pi_*(\nu) = 0$ and $\nu_x = \frac{\partial}{\partial t} + A \frac{\partial}{\partial z}$ in coordinates. $\pi_*(\frac{\partial}{\partial t}) = \frac{\partial z}{\partial t} \frac{\partial}{\partial z} + \frac{\partial \bar{z}}{\partial t} \frac{\partial}{\partial \bar{z}}$. Then at $t = 0$ we have

$$\pi_*(\nu) = \frac{\partial z}{\partial t} \frac{\partial}{\partial z} + \frac{\partial \bar{z}}{\partial t} \frac{\partial}{\partial \bar{z}} + A \frac{\partial}{\partial z} = 0$$

This shows $A = -\frac{\partial z}{\partial t} \frac{\partial}{\partial z}$ and $\frac{\partial \bar{z}}{\partial t} = 0$ which implies that π is holomorphic in t at $t = 0$. This is surprising as we don't assume the holomorphicity a priori, but we get it on a process of doing something irrelevant.

We can also prove it simply by chain rule. As $\pi(z(x, t), t) = x$, applying $\frac{\partial}{\partial t}$ on both sides and using the holomorphicity of z with respect to t and that of π with respect to z at $t = 0$, we can observe the fact immediately.